Lecture 11/12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

February 10 - 12, 2026

1 Overview

[cite_start]The goal is to determine the probability density function (PDF) and cumulative distribution function (CDF) of a new random variable Y, given that the PDF $f_X(x)$ and CDF $F_X(x)$ of an initial random variable X are known a priori[cite: 520]. [cite_start]The lecture covers transformations of single random variables, $Y = g(X)$, and functions of two random variables, such as $Z = g(X, Y)$, focusing on cases like $Z = X+Y$, $Z = X-Y$, and $Z = X/Y$[cite: 523, 535, 540, 630, 730, 742].

[cite_start]The outline of the concepts discussed includes[cite: 546]:

- **Transformation of Random Variables:** Learning techniques for random variable transformations, specifically dealing with monotonically increasing (+ve) and monotonically decreasing (-ve) functions[cite: 548, 550, 552]. [cite_start]
- **Function of Two Random Variables:** Joint transformations and derived distributions[cite: 553, 556]. [cite_start]
- **Illustrative Example:** Detailed derivation for the case $Z = X + Y$[cite: 558, 559].

2 Transformation of a Single Random Variable

[cite_start]To find the distribution of $Y = g(X)$ where $g(x)$ is a monotonic function, we use a structured step-by-step approach[cite: 548, 566].

2.1 Monotonically Increasing Function

[cite_start]If $y = g(x)$ is a monotonically increasing function, we apply the following steps[cite: 548, 550]:

- **Step 1: Express the CDF of Y.** The CDF is defined as $F_Y(y) = \Pr(Y \leq y)$[cite: 570, 579]. [cite_start]Substituting $g(X)$ for Y, we get $F_Y(y) = \Pr(g(X) \leq y)$[cite: 579]. [cite_start]
- Taking the inverse function, $x = g^{-1}(y)$, this becomes $F_Y(y) = \Pr(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$[cite: 580]. [cite_start]
- **Step 2: Differentiate with respect to Y.** The PDF is the derivative of the CDF[cite: 569, 571].

  $$f_Y(y) = \frac{d}{dy}[F_X(g^{-1}(y))]$$

  [cite_start]
- Applying the chain rule, this yields $f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}[g^{-1}(y)]$[cite: 573]. [cite_start]

- **Step 3: Change limits.** Ensure the domain of X is properly mapped to the domain of Y[cite: 591].

2.2 Monotonically Decreasing Function

[cite_start]If $y = g(x)$ is a monotonically decreasing function, the inequality direction changes[cite: 586, 589]:

- $F_Y(y) = \Pr(Y \leq y) = \Pr(X \geq g^{-1}(y))$[cite: 587, 589]. [cite_start]
- This can be expressed in terms of the CDF of X as $F_Y(y) = 1 - F_X(g^{-1}(y))$[cite: 589]. [cite_start]
- Differentiating with respect to y gives $f_Y(y) = 0 - f_X(g^{-1}(y))\frac{d}{dy}[g^{-1}(y)]$[cite: 590]. [cite_start]
- Because $\frac{d}{dy}[g^{-1}(y)]$ is negative for a decreasing function, we can write the general PDF for both cases using the absolute value[cite: 594]:

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|_{x=g^{-1}(y)}}$$

2.3 Example: Sine Transformation

[cite_start]Let X be a uniformly distributed random variable over the interval $(-1, 1)$[cite: 599]. The PDF of X is: [cite_start]

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

[cite: 601]

[cite_start]We want to find the PDF of $Y = \sin(\frac{\pi x}{2})$[cite: 602].

- First, find the inverse function: $x = \frac{2}{\pi}\sin^{-1}(y)$[cite: 603].
- Find the derivative $\frac{dx}{dy}$: [cite_start]

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

  [cite: 604] [cite_start]

- Determine the new limits: When $x = -1$, $y = \sin(-\frac{\pi}{2}) = -1$[cite: 607]. [cite_start]When $x = 1$, $y = 1$[cite: 608]. [cite_start]Thus, the bounds for Y are $-1 < y < 1$[cite: 612].
- Apply the transformation formula to find $f_Y(y)$:

$$f_Y(y) = f_X(x) \cdot \left|[cite_start]\frac{dx}{dy}\right|$$

  [cite: 609] [cite_start]

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}} = \frac{1}{\pi\sqrt{1-y^2}}$$

  [cite: 611]

3 Function of Two Random Variables

[cite_start]We are interested in finding the PDF $f_Z(z)$ for functions of two random variables, such as $Z = g(X, Y)$, where X and Y are independent random variables[cite: 617, 618, 620].

3.1 Derivation for $Z = X + Y$

[cite_start]To find the PDF of $Z = X+Y$, we first construct its CDF[cite: 630, 640]: [cite_start]

$$F_Z(z) = \Pr(Z \leq z) = \Pr(X + Y \leq z)$$

[cite: 640] [cite_start]This inequality defines a region where $x \leq z-y$[cite: 634]. [cite_start]The CDF is evaluated by integrating the joint PDF $f_{XY}(x, y)$ over this region[cite: 642]: [cite_start]

$$F_Z(z) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{z-y} f_{XY}(x, y)dxdy$$

[cite: 642]

[cite_start]To find $f_Z(z) = \frac{d}{dz}[F_Z(z)]$, we use the Leibnitz rule for differentiating under the integral sign[cite: 644, 650]. [cite_start]The Leibnitz rule states that for $G(x) = \int_{a(x)}^{b(x)} h(x, y)dy$[cite: 651]: [cite_start]

$$\frac{d}{dx}[G(x)] = \frac{d}{dx}b(x) \cdot h(x, b(x)) - \frac{d}{dx}a(x) \cdot h(x, a(x)) + \int_{a(x)}^{c} \frac{\partial}{\partial x}h(x, y)dy$$

[cite: 652]

[cite_start]Applying this to our expression for $F_Z(z)$[cite: 661, 666]:

$$f_Z(z) = \int_{-\infty}^{\infty} \left[\frac{d}{dz}(z - y) \cdot f_{XY}(z - y, y) - \frac{d}{dz}(-\infty)f_{XY}(-\infty, y) + \int_{-\infty}^{z-y} \frac{\partial}{\partial z} f_{XY}(x, y)dx \right] dy$$

[cite_start]This simplifies to the convolution integral[cite: 662, 667]: [cite_start]

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)dx$$

[cite: 667] [cite_start]Since X and Y are independent, $f_{XY}(x, y) = f_X(x)f_Y(y)$, yielding[cite: 673]: [cite_start]

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x)dx$$

[cite: 673]

3.2 Examples for $Z = X + Y$

Example 1: Sum of Independent Normal Random Variables [cite_start]Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$ be independent random variables[cite: 621]. [cite_start]We want to prove that $Z = X + Y \sim \mathcal{N}(0, 2)$[cite: 622]. [cite_start]The PDFs are $f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ and $f_Y(y) = \frac{1}{\sqrt{2\pi}}e^{-y^2/2}$[cite: 679, 687]. [cite_start]Applying the convolution integral[cite: 688]: [cite_start]

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}}e^{-(z-x)^2/2}dx$$

[cite: 688] [cite_start]Combining the exponents[cite: 688]: [cite_start]

$$-\frac{x^2}{2} - \frac{(z - x)^2}{2} = -\frac{2x^2 - 2zx + z^2}{2}$$

[cite: 688] [cite_start]Completing the square for the terms involving x yields[cite: 693]: [cite_start]

$$\frac{-\left(\sqrt{2}x - \frac{z}{\sqrt{2}}\right)^2 + \frac{z^2}{2}}{2}$$

[cite: 693] [cite_start]Factoring out the terms constant with respect to x and solving the integral gives[cite: 696]: [cite_start]

$$f_Z(z) = \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-\frac{z^2}{2(2)}} = \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/2\sigma^2}$$

[cite: 696] [cite_start]Here, $\sigma^2 = 2$, which proves that $Z \sim \mathcal{N}(0, 2)$[cite: 696].

Example 2: Sum of Independent Exponential Random Variables [cite_start]Let X and Y be exponential random variables with parameter λ[cite: 702, 708]. [cite_start]

$$f_X(x) = \lambda e^{-\lambda x}, \quad f_Y(y) = \lambda e^{-\lambda y}$$

[cite: 708] [cite_start]For $Z = X + Y$, the PDF is[cite: 708]: [cite_start]

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x) dx$$

[cite: 708] [cite_start]Substituting the PDFs, noting the limits governed by the step functions $u(x)$ and $u(z - x)$[cite: 710, 718]: [cite_start]

$$f_Z(z) = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} u(x) \cdot \lambda e^{-\lambda(z-x)} dx$$

[cite: 710]

3.3 Other Operations

Difference: $Z = X - Y$ [cite_start]For $Z = X - Y$[cite: 730]:

- The CDF is $F_Z(z) = \Pr(Z \leq z) = \Pr(X - Y \leq z)$[cite: 736]. [cite_start]
- This corresponds to the region $x \leq z + y$[cite: 734]. [cite_start]
- The CDF is formulated as $F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y) dx dy$[cite: 737]. [cite_start]
- The PDF $f_Z(z) = \frac{d}{dz}[F_Z(z)]$ is derived using the same procedure as the sum $Z = X + Y$[cite: 737].

Quotient: $Z = \frac{X}{Y}$ [cite_start]For the ratio $Z = \frac{X}{Y}$[cite: 742]:

- The CDF is $F_Z(z) = \Pr(Z \leq z) = \Pr(\frac{X}{Y} \leq z)$[cite: 742].